



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

EC3223-1 - REAL TIME OPERATING SYSTEMS USING QNX
(For EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which file describes the layout and contents of a QNX boot image?
A) buildfile
B) config.ini
C) bootmap.txt
D) startup.cfg
2. The buildfile used in QNX boot image creation contains:
A) Instructions for assembling the image
B) Source code for the kernel
C) User credentials
D) Debugging logs
3. In QNX, a resource manager is responsible for:
A) Scheduling threads
B) Handling I/O requests and device interactions
C) Scheduling threads
D) Encrypting files
4. Resource managers in QNX operate in:
A) Kernel space
B) User space
C) BIOS layer
D) Hardware abstraction layer
5. Which function is used to register a resource manager in QNX?
A) resmgrattach
B) registemgr
C) attachresource
D) mgrbind
6. A resource manager in QNX can be used to:
A) Encrypt memory
B) Compile applications
C) Monitor CPU temperature
D) Create custom device interfaces
7. The main advantage of resource managers in QNX is:
A) Modular and flexible I/O handling
B) Faster boot times
C) Reduced power consumption
D) Enhanced graphics performance
8. Which function is used to send a message in QNX IPC?
A) MsgSend
B) SendMsg
C) ipc_send
D) postMessage
9. Resource managers in QNX are used to:
A) Encrypt data
B) Manage user accounts
C) Handle I/O operations and device interactions
D) Monitor CPU temperature
10. Which function is used to create a new process in QNX?
A) start()
B) exec()
C) run()
D) spawn()

11. To terminate a process in QNX, you can use:
 A) stop() B) exit()
 C) kill_thread() D) delete()
12. Memory protection in QNX ensures:
 A) One process cannot access another's memory B) All processes share memory
 C) Memory is encrypted D) Memory is compressed
13. Which function is used to create a thread in QNX?
 A) newthread B) threadstart
 C) createthread D) pthreadcreate
14. To terminate a thread in QNX, use:
 A) pthreadexit B) threadkill
 C) stopthread D) removethread
15. Which synchronization primitive prevents race conditions?
 A) Mutex B) Timer
 C) Signal D) Pipe
16. Which IPC method is most suitable for real-time control signals?
 A) Message passing B) Shared memory
 C) Pulses D) File I/O
17. Which QNX function is used to create a shared memory object?
 A) memalloc B) createmem
 C) shmopen D) openshared
18. Which IPC method in QNX is best for small, fast notifications?
 A) Shared memory B) Pulses
 C) Message passing D) Signals
19. IO-mapped and memory-mapped IO are techniques used for:
 A) Accessing hardware devices B) Managing virtual memory
 C) Scheduling processes D) Handling file systems
20. Interrupt handling in QNX involves:
 A) Attaching an ISR to a hardware event B) Using semaphores for synchronization
 C) Creating threads for each interrupt D) Polling devices continuously

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Describe how the spawn() function works in QNX. How is it different from fork() and exec() used in traditional UNIX systems?	8	L*2	1	1
	b) Explain the message-passing mechanism in QNX. How do the MsgSend(), MsgReceive(), and MsgReply() system calls facilitate synchronous communication between client and server processes?	8	L3	2	1
2.	a) Explain the purpose and function of Process ID (PID) and Parent Process ID (PPID) in process management. How does the QNX OS use these identifiers to maintain process hierarchy?	8	L2	1	1
	b) Explain how mutexes, semaphores, and condition variables are implemented in QNX for synchronization. Describe how they prevent race conditions and deadlocks in multi-threaded applications.	8	L3	2	1
3.	a) Explain how thread states (READY, RUNNING, BLOCKED, etc.) are managed in QNX. Give examples of transitions between these states during synchronization or IPC operations.	8	L3	1	1

- b) Discuss how QNX handles process priorities and scheduling policies. How does the system ensure that real-time processes meet their deadlines?

8 L2 2 1

Unit – II

4. a) Describe message passing, pulses, and shared memory mechanisms, explaining how each is used for communication between processes and threads in a microkernel environment.
b) Describe how QNX allows user-space processes to access hardware I/O ports and registers.

8 L3 3 1

8 L2 4 1

5. a) Discuss how message passing provides safety and synchronization, how shared memory offers speed, and how pulses are useful for lightweight event notifications. Include suitable examples.
b) What is DMA-safe memory allocation, and why is it important in QNX?

8 L3 3 1

8 L2 4 1

6. a) Explain the steps involved in implementing shared memory communication between two processes in QNX.
b) Differentiate between I/O-mapped I/O and memory-mapped I/O in embedded systems.

8 L2 3 1

8 L2 4 1

Unit – III

7. a) Describe the process of building a QNX boot image using the QNX Momentics IDE or command-line tools.
b) Explain how developers can modify the startup code, add or remove drivers, and edit the buildfile to tailor the image for specific hardware.

8 L2 5 1

8 L2 5 1

8. a) Explain the purpose of startup code, kernel and device drivers in QNX OS.
b) What are resource managers in QNX, and how are they implemented?

8 L2 5 1

8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
